higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

AUTOMOTIVE REPAIR AND MAINTENANCE
NQF LEVEL 4

22 NOVEMBER 2019

This marking guideline consists of 5 pages.

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SECTION A

QUESTION 1

1.1 A
1.2 C
1.3 D
1.4 D
1.5 D
1.6 B
1.7 C
1.8 A
1.9 C
1.10 A

(10 × 1) **[10]**

QUESTION 2

2.1 False
2.2 True
2.3 False
2.4 True
2.5 False

(5 × 1) **[5]**

QUESTION 3

3.1 F
3.2 D
3.3 A
3.4 M
3.5 I
3.6 L
3.7 H
3.8 B
3.9 E
3.10 G

(10 × 1) **[10]**

QUESTION 4

4.1 slower
4.2 time
4.3 distance
4.4 planetary
4.5 cross shaft

(5 × 1) **[5]**

TOTAL SECTION A: 30

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SECTION B

QUESTION 5

5.1 5.1.1 Splash method
5.1.2 Reduces friction/reduces heat
5.1.3 SAE 80 W/90 gear oil

(3 × 1) (3)

5.2 It consists of an ordinary pinion gear and crown wheel giving the primary reduction.✓ The pinion gear mounted on the cross shaft of the crown wheel✓ meshes with the helical drive gear to form the second reduction.✓ (3)

5.3
- Do not handle refrigerant near open flames.
- Always wear eye protection.
- Do not spill refrigerant on the skin.
- Ensure the system is fully discharged before removing any pipes.
- Never heat the service can.
- Remove all used components and dispose of it in appropriate containers.
- Discharge systems only into a recovery unit. (Any 5 × 1) (5)

5.4
- Expansion-valve system
- Fixed orifice-tube system (2)

5.5
- Apply the handbrake and press down the brake pedal.
- Start the engine and put the transmission in drive.
- Slowly increase the engine r/min to the specifications.
- At the specified r/min the engine should stall. (4)

5.6
- Reduces driver fatigue
- Saves the driver effort of selecting gears and depressing a clutch
- Safer to drive due to the driver having both hands on the steering wheel
- Provides a smoother pull away
- Reduces fuel consumption
- Extends engine life due to lower revolutions (Any 3 × 1) (3)

[20]

QUESTION 6

6.1 6.1.1
- EIS ignites the air/fuel mixture in the combustion chamber.
- EFI delivers fuel and it occurs before the EIS ignites the air/fuel. (2)

6.1.2
- Do not disconnect the ECU harness with the ignition ON.
- Always release the fuel pressure before disconnecting EFI lines.
- Do not connect direct battery voltage to the injectors as this will damage the resistors.
- Always use a high-impendence meter. (4)

6.1.3
- Connect the pressure gauge to the fuel rail.
- Disable the ignition system to prevent the engine from starting.
- Crank the engine and check the fuel pressure.
- If pressure is zero there is a blockage in the fuel lines. (4)

6.2 6.2.1 Operating pressure is on the delivery✓ side of the fuel pump, while✓ retention pressure is still in the system✓after the engine has been switched off.✓ (2 + 2) (4)

6.2.2 A mechanical electronic injector is mechanically actuated✓ but electronically controlled,✓ while a hydraulic energy injector uses hydraulic energy✓ from pressurised engine lubrication.✓ (2 + 2) (4)

6.3
- Fit a flowmeter to the delivery side of the fuel pump.
- Start the engine and check the fuel flow.
- Check the reading and compare it with the specifications. (3)

6.4
- Connect the ignition system.
- Start the engine and check for fuel line leaks.
- While the engine is running listen for any air leaks.
- See if the engine idles unevenly. (Any relevant method) (4)

[25]

QUESTION 7

7.1
- Adjustable screw
- Rack-guide bushing
- Rack teeth
- Pinion teeth
- Rack-trust spring (Any 3 × 1) (3)

7.2 An electric hydraulic steering pump will still be operative even if the engine is switched off, but electrical power is on, while an engine-driven steering pump will not be operative if the engine is switched off. (2)

7.3
- Fuel efficiency
- No belt-driven accessories required
- No high-pressure hydraulic hoses required (3)

7.4 7.4.1
- Retrieve fault codes from various control units
- Convert computer pulses into digital display

7.4.2
- Identify different wave forms of various components
- Eliminate damaging the electronic system and wiring

(2 × 2) (4)

7.5 7.5.1 Kinetic friction occurs when the wheels lock and slide when brakes are applied, while static friction occurs when the wheels continue to rotate while braking.

7.5.2 Braking force is the force between the tyres and the road surface, while accelerating force is a result of the amount of torque applied.

7.5.3 TCS is a secondary function of the ABS that detects one wheel spinning faster and then applies brakes to it, while ESC helps to steer the vehicle where the driver intends to go. (3 × 2) (6)

7.6
- Superior fuel efficiency
- Massive reduction in CO (2)

7.7
- Safety belt
- SRS indicator light
- Airbags
- Impact sensors (4)

7.8 True (1)
[25]

TOTAL SECTION B: 70
GRAND TOTAL: 100

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